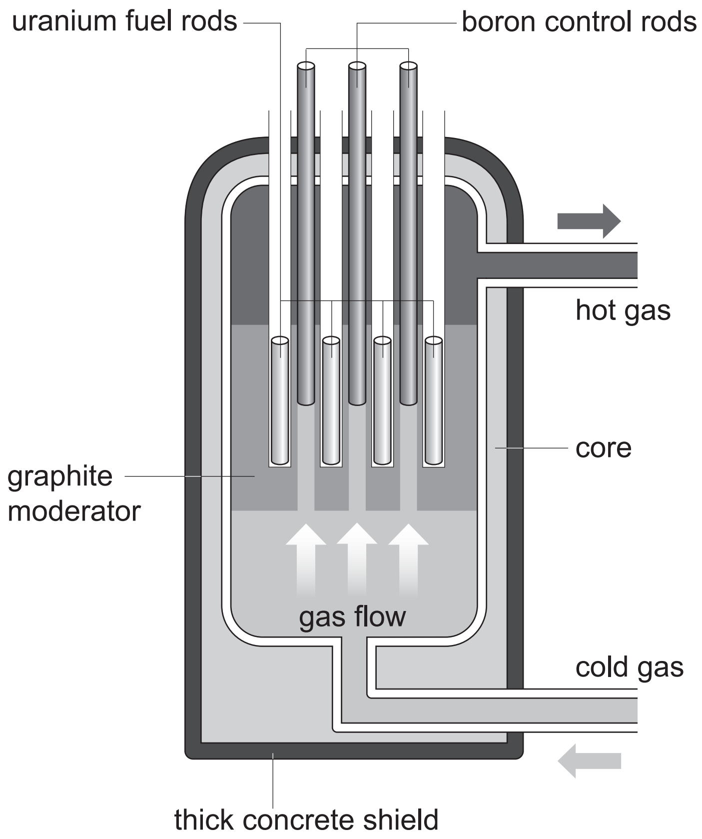
Examiner only

6. Nuclear fission is a nuclear reaction that releases energy.
The diagram shows a nuclear fission reactor.



(a) In each fission reaction, on average three neutrons are released.

Explain how this can lead to an uncontrolled chain reaction **and** how the features of the reactor allow a controlled chain reaction to be achieved. [6 QER]

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(b) Another nuclear reaction is nuclear fusion.

One fusion reaction, which could provide energy on Earth in the future, involves two isotopes of hydrogen, H, deuterium and tritium.

Deuterium and tritium fuse to produce an isotope of helium, He, and one neutron.

(i) Use the information from the table below to complete the nuclear equation for this reaction. [3]

Isotope	Number of protons	Number of neutrons
deuterium	1	1
tritium	1	2

$$^{\cdots}_{\cdots}\mathrm{H} + {}^{\cdots}_{\cdots}\mathrm{H} \longrightarrow {}^{\cdots}_{\cdots}\mathrm{He} + {}^{\cdots}_{\cdots}\mathrm{n}$$

(ii) The main product of this reaction is helium.

Suggest why, if it can be achieved on Earth, nuclear fusion may be better for energy production than nuclear fission. [1]

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